NAME: _____

Standing Wave and Resonance — second case

WHERE WE LEFT OFF

Last class we worked the Tacoma Narrows case and finished at the Reasoning step with **resonance leads to amplification of vibrational amplitude**. The second case on your handout is the opera singer. It is the same four moves — describe, Claim, Evidence, Reasoning — on an object whose confined space is a wine glass rather than a bridge deck. **These are items 6–10 of your Note 1 handout.** Use this sheet if you do not have that copy with you.

Keep the same four things in view while you watch. They are the ones from Note 1:

i) The traveling waves generated from a source are the **driving waves** that excite the object and whose frequencies are the so-called **driving frequencies**.

ii) The condition for exciting the object's natural frequency creating resonance.

iii) When the resonance condition is met, what happens to the wave's amplitude in the object?

iv) What happens when the resonance frequency of the object is inadvertently excited? Is the resonance phenomenon a good thing from a structural engineering perspective?

Instructions and Questions

6 Type "Failure is not an option for this Opera Singer" into the YouTube search bar and watch the video.

youtube.com/watch?v=uDyri-KPOAc

7 Briefly describe what happened.

8 **Claim** (Answer the question, providing hypothesis)
"How did the opera singer break the glass?"

9 **Evidence** (Describe your observations or cite any relevant data.)

10 **Reasoning** (Providing scientific rationale for your claims, backing them up with your observations. **Incorporate i) – iv) in your answer**.)

Finally, find out why Millenium Bridge in London, UK, was shut down for many months and what the fixes were.

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